Using these vectors, the non-parametric form of vector equation of the plane passing through the given three non-collinear points can be equivalently written as

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 0$$

which is the Cartesian equation of the plane passing through three non-collinear points.

6.8.5 Equation of a plane passing through a given point and parallel to two given non-parallel vectors.

(a) Parametric form of vector equation

Consider a plane passing through a given point A with position vector $\vec{a}$ and parallel to two given non-parallel vectors $\vec{b}$ and $\vec{c}$. If $\vec{r}$ is the position vector of an arbitrary point P on the plane, then the vectors $(\vec{r}-\vec{a}), \vec{b}$ and $\vec{c}$ are coplanar. So, $(\vec{r}-\vec{a})$ lies in the plane containing $\vec{b}$ and $\vec{c}$. Then, there exists scalars $s, t \in \mathbb{R}$ such that $\vec{r}-\vec{a} = s\vec{b}+t\vec{c}$ which implies

$$\vec{r} = \vec{a}+s\vec{b}+t\vec{c}, \text{ where } s, t \in \mathbb{R} \qquad \ldots (1)$$

This is the parametric form of vector equation of the plane passing through a given point and parallel to two given non-parallel vectors .

(b) Non-parametric form of vector equation

Equation (1) can be equivalently written as

$$(\vec{r}-\vec{a})\cdot(\vec{b}\times\vec{c}) = 0 \qquad \ldots (2)$$

which is the non-parametric form of vector equation of the plane passing through a given point and parallel to two given non-parallel vectors .

(c) Cartesian form of equation

If $\vec{a} = x_1\hat{i}+y_1\hat{j}+z_1\hat{k}$, $\vec{b} = b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$, $\vec{c} = c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$ and $\vec{r} = x\hat{i}+y\hat{j}+z\hat{k}$, then the equation (2) is equivalent to

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

This is the Cartesian equation of the plane passing through a given point and parallel to two given non-parallel vectors.

6.8.6 Equation of a plane passing through two given distinct points and is parallel to a non-zero vector

(a) Parametric form of vector equation

The parametric form of vector equation of the plane passing through two given distinct points A and B with position vectors $\vec{a}$ and $\vec{b}$, and parallel to a non-zero vector $\vec{c}$ is

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